Quiz 1: Data Science Foundations and Python Basics

Introduction to Data Science · Weeks 1–2 · 15 minutes · 10 marks · closed notes, no phones

Name: _____ Roll no.: _____ Date: _____

Section A: Multiple choice (8 marks)

Choose the single best answer. One mark each.

1. Which statement best describes Data Science?

A. Training machine-learning models on large datasets.

B. Using data to answer questions and support decisions, through a process that runs from the question to communicating the result.

C. Writing Python programs that read files.

D. Building and maintaining databases.

2. A company wants to collect millions of orders, store them reliably and prepare them for analysts. Which role mainly does this work?

A. Data Engineer

B. Statistician

C. Machine Learning researcher

D. Presentation designer

3. In a table with columns `order_id`, `customer_id` and `amount`, each row is one order. Customer C1 placed three orders. How many rows belong to C1?

A. 1

B. 3

C. It depends on the month.

D. 0

4. Which of these is an example of unstructured data?

A. A CSV file of daily sales.

B. A JSON record with named fields.

C. A customer review written as free text.

D. A table of ages.

5. What does `'24' == 24` evaluate to in Python?

A. True

B. False

C. It raises an error.

D. 24

6. Which Python container is best for one CSV row with named fields, such as a city and a quantity?

A. list

B. tuple

C. dict

D. set

7. For `amounts = [1200, 450, 3000, 80, 950, 5000]`, what does `amounts[1:4]` return?

A. `[450, 3000, 80, 950]`

B. `[450, 3000, 80]`

C. `[1200, 450, 3000, 80]`

D. `[3000, 80, 950]`

8. In a notebook, a variable `x` still works after you delete the cell that created it. What is the best explanation?

A. Python is broken.

B. The notebook session still remembers values from cells that were already run.

C. The variable was saved to the hard disk.

D. Cells always run from top to bottom automatically.

Section B: Short conceptual questions (2 marks)

Answer in two to three sentences. One mark each.

9. A program reads 1,000 rows. Three of them contain the text `N/A` in a column that should hold numbers. Describe what a good program should do with those rows, and say why.

10. Why should you ask “what does one row represent?” before combining two tables? Give one example of what can go wrong.

End of quiz.